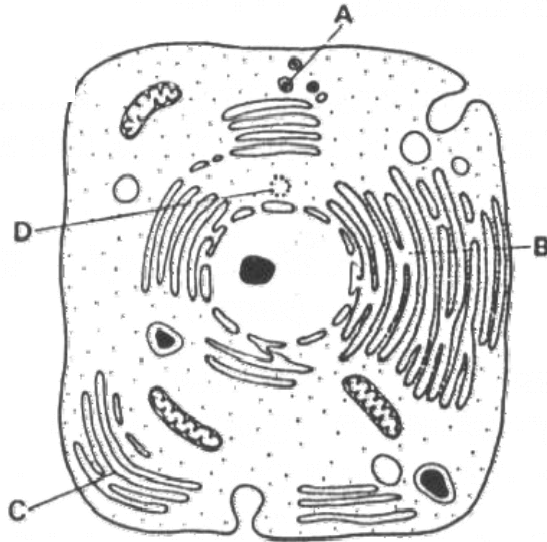


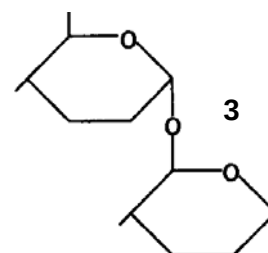
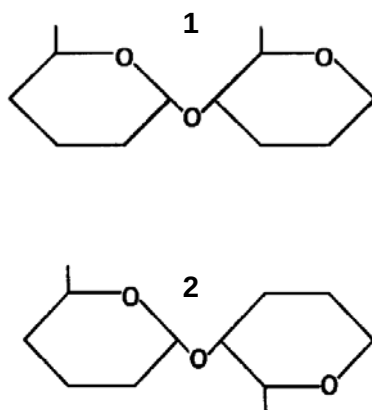
- 1 The diagram below shows some of the structures in an animal cell.

Which one of the following structures A, B, C, D would be involved in the final secretion of digestive enzymes from this cell?



- 2 The R group of the amino acid serine is $-\text{CH}_2\text{OH}$. The R group of the amino acid alanine is $-\text{CH}_3$. Where would you expect to find these amino acids in a globular protein in aqueous solution?
- A Serine would be in the interior, and alanine would be on the exterior of the globular protein.
 - B Alanine would be in the interior, and serine would be on the exterior of the globular protein.
 - C Both serine and alanine would be in the interior of the globular protein.
 - D Both serine and alanine would be in the interior and on the exterior of the globular protein.

- 3 The diagrams show different types of bonds found in polysaccharides.



Which type or types of bonds is/are found in amylose?

- A 1 only
- B 2 only
- C 1 and 3
- D 2 and 3

- 4 Three pure substances are analyzed. The table shows the elements that each contains.

	C	H	O	N	P	S
X	+	+	+	+	-	-
Y	+	+	+	-	-	-
Z	+	+	+	+	-	+

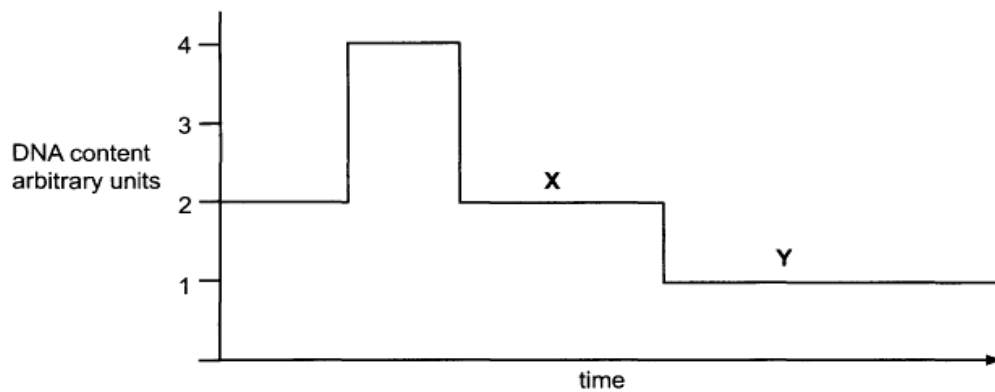
Which combination could be correct?

	X	Y	Z
A	adenine	carbohydrate	fat
B	adenine	fat	an amino acid
C	an amino acid	fat	carbohydrate
D	fat	carbohydrate	an amino acid

- 5 Which statement is true for a competitive inhibitor of enzyme activity, but not for a non-competitive inhibitor?

- A Inhibitor molecules are similar to normal substrate molecules in size and shape.
- B Inhibition cannot be reduced by increasing the concentration of substrate molecules.
- C Small numbers of inhibitor molecules are needed for effective enzyme inhibition.
- D The inhibitor binds to the enzyme molecule at a position away from the active site.

- 6 The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.



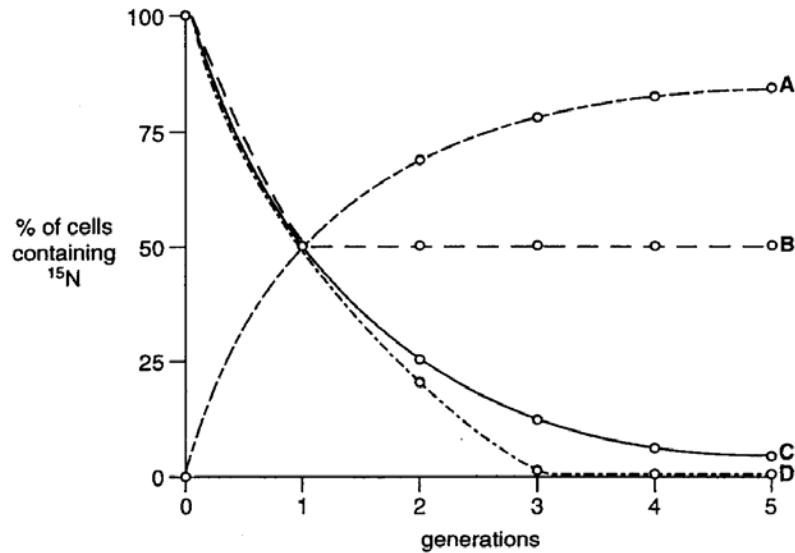
Which combination is correct?

	Ploidy of cell at end of X	Ploidy of cell at end of Y
A	diploid	diploid
B	diploid	haploid
C	haploid	haploid
D	haploid	diploid

- 7 Which event does **NOT** occur in meiosis?
- A Recombination of alleles of different genes on chromatids.
 - B Attachment of microtubules to kinetochore proteins.
 - C Synthesis of new DNA strands by semi-conservative replication.
 - D Chromatin fibres forming looped domains.

- 8 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labeled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative?



- 9 What is an accurate description of the coding of protein structure by DNA?
- A a degenerate, non-overlapping code for the primary structure, determining the location of the folding sites involved in tertiary structure.
 - B a code, with several triplets for each amino acid, determining the shape of the α -helix for the secondary structure.
 - C 64 different complementary codons used in transcription to make the three-dimensional shape of the tertiary structure.
 - D amino acids coded for by three consecutive bases, used with tRNA anticodons in translation to determine the quaternary structure.

- 10 The table shows the mRNA codons for 11 different amino acids.

Amino acid	mRNA codon
Ala	GCG
Glu	GAG
His	CAC
Leu	CUG CUC

Amino acid	mRNA codon
Lys	AAG
Pro	CCU
Thr	ACU
Val	GUG

Amino acid	mRNA codon
Arg	CGC
Phe	UUC
Gly	GGA

The first seven DNA triplets coding for a protein are shown below. These are DNA triplets from the coding strand, complementary to the mRNA.

CACGAGGCGAAGGGACACCGC

A mutation occurs when the seventeenth nucleotide in the DNA sequence is changed to G.

What is the amino acid sequence for the mutated section of DNA?

- A His Glu Ala Lys Gly Arg Arg
 - B Val Leu Arg Phe Pro Ala Ala
 - C His Glu Ala Lys Gly His Arg
 - D Val Leu Arg Phe Pro Val Ala
- 11 In the pathway of lactose utilization by *E. coli*, when the lac operon is induced the
- A lac repressor is active and transcription of the lac operon does not occur
 - B lac repressor is inactive and the lac operon is transcribed
 - C lac repressor is inactive and transcription of the lac operon does not occur
 - D lac repressor is active and the lac operon is transcribed

- 12 Which of the following pairs of statements is true of generalized and specialized transduction?

	Generalized	Specialized
I	Transfers any bacterial DNA	Transfers a specific set of bacterial genes
II	Contains a hybrid chromosome in its capsid	Contains only bacterial chromosome in its capsid
III	The host cell will die	The host cell may die
IV	Viral genome is not transcribed	Viral genome is transcribed
V	Viral DNA is integrated into the host chromosome	Viral DNA is not integrated into the host chromosome
VI	Viral DNA is replicated by host cell machinery	Viral DNA is replicated by binary fission

- A I, IV and V
 B II, III and VI
 C III, IV and V
 D I, III and VI
- 13 Which best describes the correct order (from earliest to latest) in the control of gene expression in protein synthesis?
- A Polyribosome, promoter, spliceosome, primary transcript
 B Spliceosome, promoter, primary transcript, polyribosome
 C Promoter, primary transcript, spliceosome, polyribosome
 D Primary transcript, spliceosome, promoter, polyribosome
- 14 A mutation that renders the product of a regulatory gene non-functional for an inducible operon will result in
- A continuous transcription of the genes of the operon.
 B irreversible binding of the repressor to the operon.
 C complete blocking of the attachment of RNA polymerase to the promoter.
 D no difference in the transcription rate.
- 15 All of the following mechanisms have been shown experimentally to contribute to the formation of cancer cells **except**
- A mutations that cause excess production of growth factors by cancer cells.
 B mutations that reduce the need for growth factors in cancer cells.
 C abnormally high energy reserves that cause cells to divide too quickly becoming cancerous.
 D viruses that carry genes that transform normal cells into cancer cells.

- 16 In bee colonies, male bees arise from unfertilized eggs while female bees arise from fertilized ones. However, females can either develop into workers or queens which have distinct phenotypes.

Which of the following best explains the phenotypic differences between the queen and the worker bees?

- A Random fusion of gametes.
- B Genetic recombination during meiosis.
- C Polygenic inheritance.
- D Environmental influence during development.

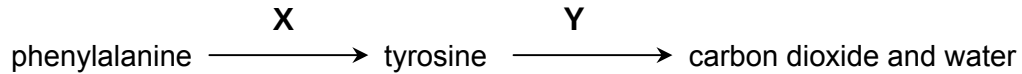
- 17 In domestic cats, the genes for yellow and black fur are an allelic pair which are sex-linked and co-dominant, the heterozygote being tortoise-shell, a mixture of yellow and black fur.

When yellow males are mated with tortoise-shell females, what are the expected ratios of phenotypes and sexes in the offspring?

- | | males | females |
|----------|--------------------|-----------------------------|
| A | 1 yellow : 1 black | 1 yellow : 1 tortoise-shell |
| B | All black | 1 yellow : 1 black |
| C | 1 yellow : 1 black | 1 black : 1 tortoise-shell |
| D | All yellow | 1 yellow : 1 black |

	X^A	Y
X^A	$X^A X^A$ Yellow female	$X^A Y$ Yellow male
X^B	$X^A X^B$ Tortoise-shell female	$X^B Y$ Black male

- 18 The following reaction sequence occurs in humans.



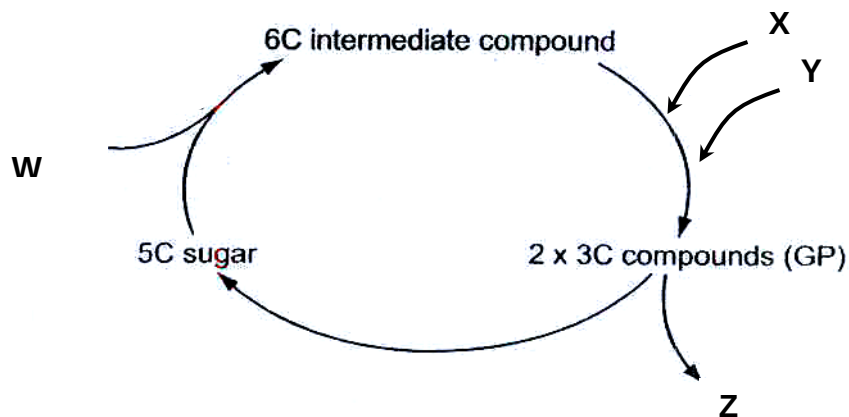
Genetic disease **P** is caused by an enzyme deficiency in step **X** and genetic disease **Q** is caused by an enzyme deficiency in step **Y**. The gene that codes for enzymes catalyzing steps **X** and **Y** are found on chromosomes 12 and 3 respectively.

An unaffected male who is carrier of disease **P** marries an unaffected female who is a carrier for disease **Q**.

What is the probability of their first son having both diseases **P** and **Q**?

- A 0.5
- B 0.25
- C 0.125
- D 0

- 19 The diagram shows some of the reactions that occur during the light-independent reactions of photosynthesis.



Identify substances **W**, **X**, **Y** and **Z**.

- | | W | X | Y | Z |
|----------|------------------|------------------|-----------------|----------------|
| A | H ₂ O | NADPH | ATP | O ₂ |
| B | ATP | H ₂ O | CO ₂ | O ₂ |

	dehydrogenase		decarboxylase	
B	pyruvate	ethanal	alcohol	ethanol
	decarboxylase		dehydrogenase	
C	pyruvate	ethanol	alcohol	ethanal
	decarboxylase		dehydrogenase	
D	pyruvate	ethanal	alcohol	ethanol
	dehydrogenase		decarboxylase	

- 22** In a plant cell, what substance enters and leaves a mitochondrion during aerobic respiration?

	enters	leaves
A	acetyl coenzyme A	carbon dioxide
B	ADP	lactate
C	glucose	reduced NADP
D	pyruvate	ATP

- 23** Homeostasis in a mammal is defined as the maintenance of a

- A** constant internal environment, providing a degree of independence from the external environment.
- B** constant internal environment, resulting in complete dependence on the external environment.
- C** variable internal environment, resulting from a variable external environment.
- D** constant internal and external environments resulting from temperature fluctuations.

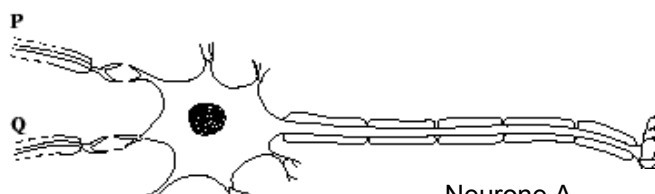
- 24** Which of the following statements BEST describes the outcome of the negative feedback loop in a homeostatic mechanism?

- A** Glucagon is released by the pancreas in response to a drop in blood glucose concentration below the normal level.
- B** The liver cells break down glycogen to glucose in response to glucagon.
- C** Blood glucose level increases to the normal level in response to the release of glucagon.
- D** Release of glucagon is stopped in response to blood glucose concentration returning to the normal level.

- 25** Which axon would transmit an action potential most rapidly?

- A** 1 mm diameter neuron lacking myelin.
- B** 1 mm diameter neuron with myelin.
- C** 2 mm diameter neuron lacking myelin.
- D** 2 mm diameter neuron with myelin

- 26** The figure below shows the effect of impulses from neurones P and Q on the production of an action potential in neurone A.

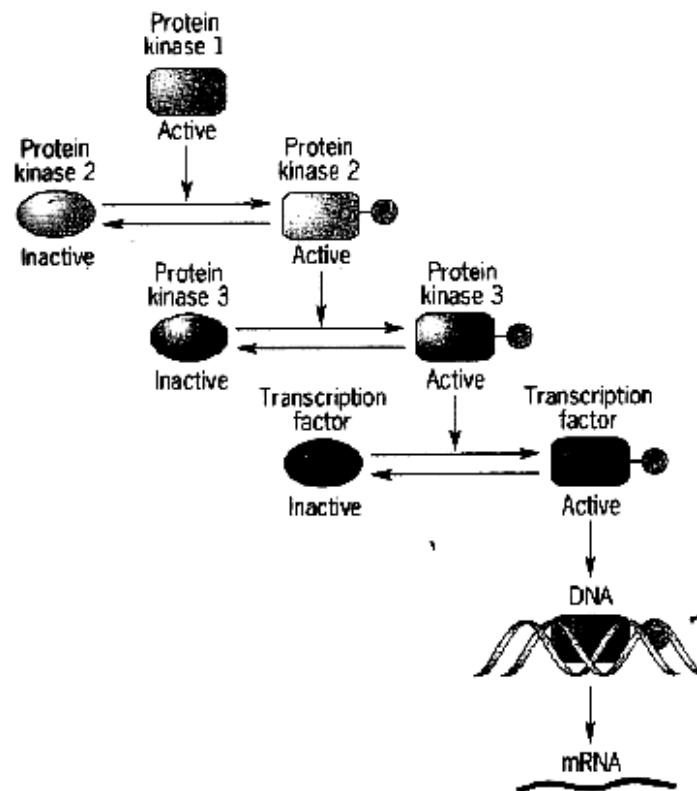


Which of the following best describes X and Y?

	X	Y
A	IPSP	EPSP
B	EPSP	IPSP
C	Spatial summation	Temporal summation
D	Temporal summation	Spatial summation

- 27** The figure below shows part of a signaling pathway in a cell. The pathway leads to the activation of a gene that codes for a molecular inhibitor of a cyclin-dependent kinase responsible for moving a cell into the S phase of the cell cycle.

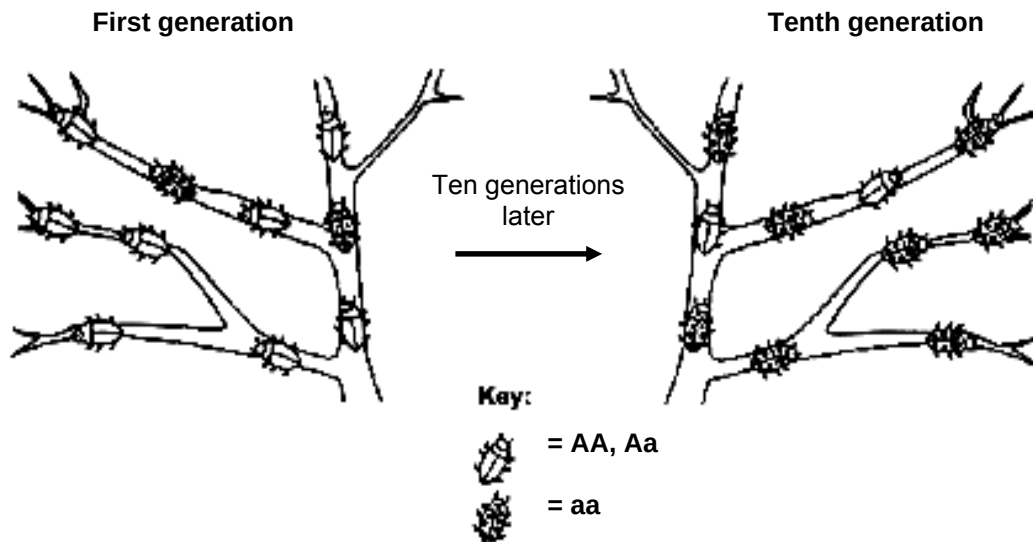
How would a mutation that results in a non-functional Protein kinase 3 affect the cell's rate of division?



- A Cell division would proceed at a rapid rate.
 B Cell division would proceed at the normal rate.
 C Cell division would proceed, but at a decreased rate.
 D Cell division would be completely inhibited.
- 28 Which of the following statements does **NOT** describe the process of signal reception by a membrane bound receptor molecule?
- A Activation of phospholipase C (PLC).
 B Phosphorylation of tyrosine residues on intracellular tail.
 C Aggregation of receptor monomers to form dimer.
 D Displacement of a protein subunit.
- 29 Cell signaling pathways always involves the binding of a ligand to a receptor. Which of the following features does **NOT** describe a ligand that binds to a receptor on cell membrane surface?
- A Comprises of amino acids
 B Complementary in shape to ligand
 C Large molecules
 D Hydrophilic in nature
- 30 Which of the following features describes the structure of a G-protein-coupled receptor?

- A Consist of α and β subunits only.
- B Consist of a single transmembrane α -helix.
- C Consist of a catalytic subunit.
- D Consist of seven transmembrane α -helices.

- 31 The diagram below illustrates the change that occurred in the frequency of phenotypes in an insect population over ten generations.



Which of the following is a probable explanation for this change?

- A Over time, there was a decrease in the adaptive value of allele **a**.
 - B Over time, there was an increase in the adaptive value of allele **a**.
 - C Over time, there was an increase in the population of this insect.
 - D Over time, there was a decrease in the mutation rate of allele **A**.
- 32 Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?
- A The fossil records.
 - B Homologous structures.
 - C Comparative embryology.
 - D Molecular comparisons of DNA and amino acid sequences.
- 33 A comparison was made between human, rabbit, mouse and chimpanzee in the
- DNA coding sequence of the β globin gene

- DNA sequence in the introns of the β globin gene
- Amino acid sequence of the β globin polypeptide

The data is shown in Table 1.

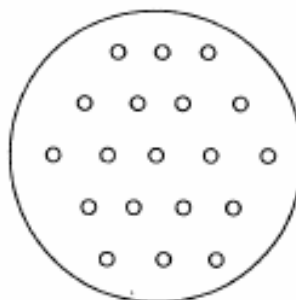
Table 1

Organisms being compared	sequence similarity (%)		
	coding DNA	intron	amino acid sequence
human β globin/chimpanzee β globin	100	98.4	100
human β globin/rabbit β globin	89.3	67	90.4
human β globin/mouse β globin	82.1	61	80.1

It is possible to conclude from this data that

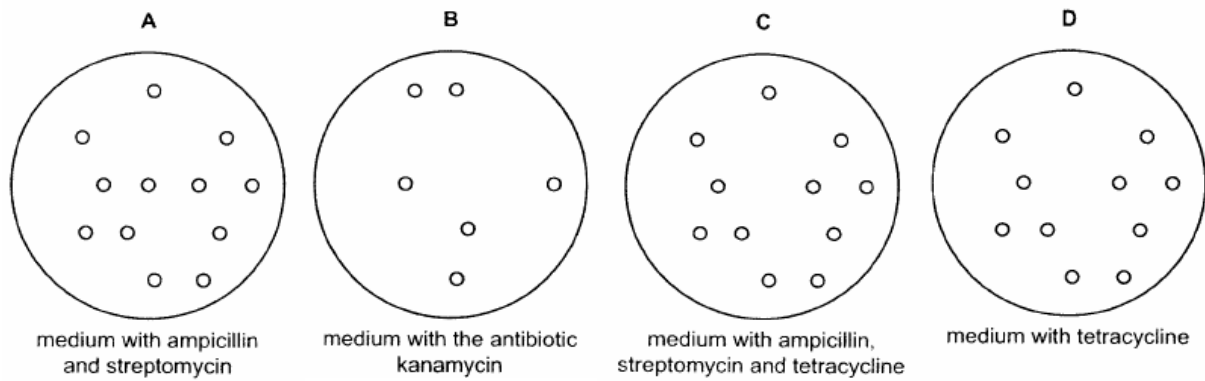
- A** a human is more closely related to a mouse than to a rabbit.
- B** the variation between chimpanzees and human occurs in a region of the β globin gene which would code for amino acids.
- C** the variation in the intron sequence between human and mouse would account for some of the differences in the amino acid sequence.
- D** the comparison between human and chimpanzee indicates that human is more closely related to chimpanzee than to rabbit and mouse.

- 34** A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria grown on a nutrient medium. The resulting master plate is shown in the diagram.

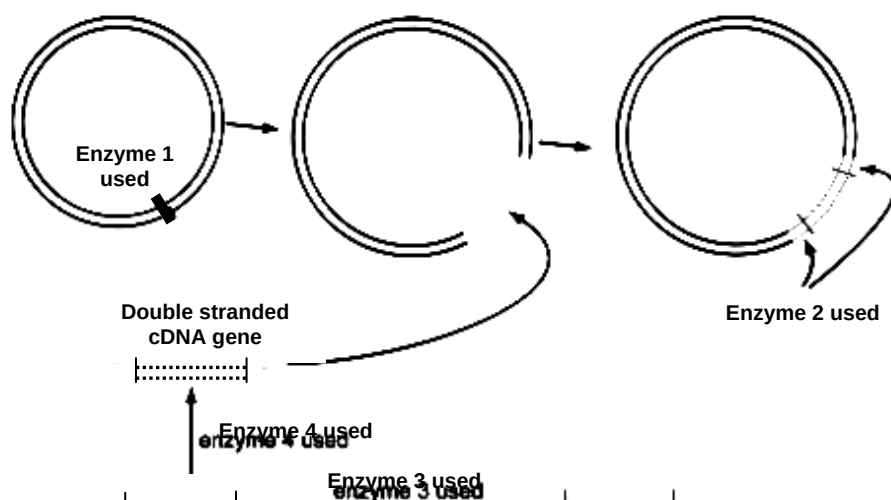


Transformed cells were selected by replica plating the bacteria colonies onto media containing various antibiotics.

Which plate contains the colony of bacteria into which the human gene has been successfully inserted?



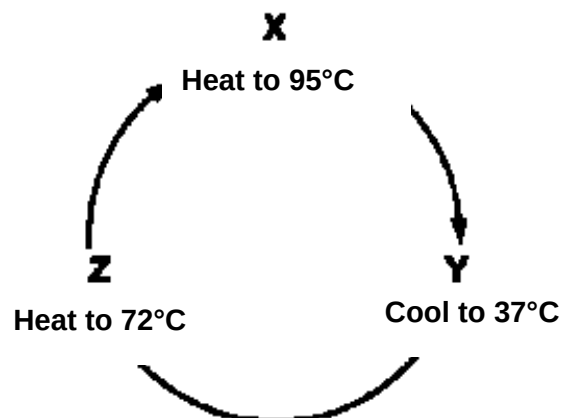
35 The diagram shows part of a process used for genetic engineering.



Which of the following shows the correct identity of the enzymes used in this part of the process?

	Enzyme			
	1	2	3	4
A	Restriction enzyme	DNA ligase	RNA polymerase	Reverse transcriptase
B	Restriction enzyme	DNA ligase	Reverse transcriptase	DNA polymerase
C	DNA ligase	Restriction enzyme	DNA polymerase	Reverse transcriptase
D	Restriction enzyme	Reverse transcriptase	RNA polymerase	DNA ligase

- 36** The diagram shows the changes in temperature of the reaction mixture during the polymerase chain reaction (PCR).



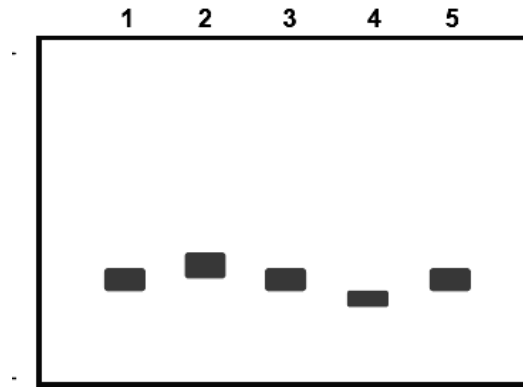
The main events during one cycle of the reaction are listed.

- 1 Binding of RNA primers
- 2 DNA synthesis
- 3 Separation of DNA strands

Which combination correctly matches each event with the temperature in the reaction mixture?

	X	Y	Z
A	1	2	3
B	2	3	1
C	3	1	2
D	3	2	1

- 37** The diagram shows the results of polyacrylamide gel electrophoresis of the same protein extracted from five different individuals.

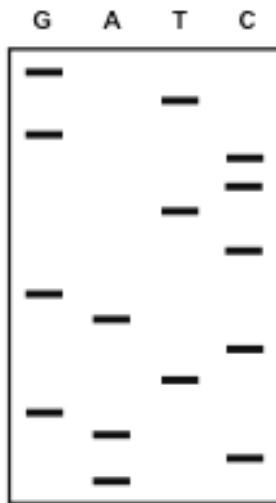


Which of the following correctly explains why the proteins extracted from individuals 2 and 4 did not travel the same distance on the gel as those extracted from individuals 1, 3 and 5?

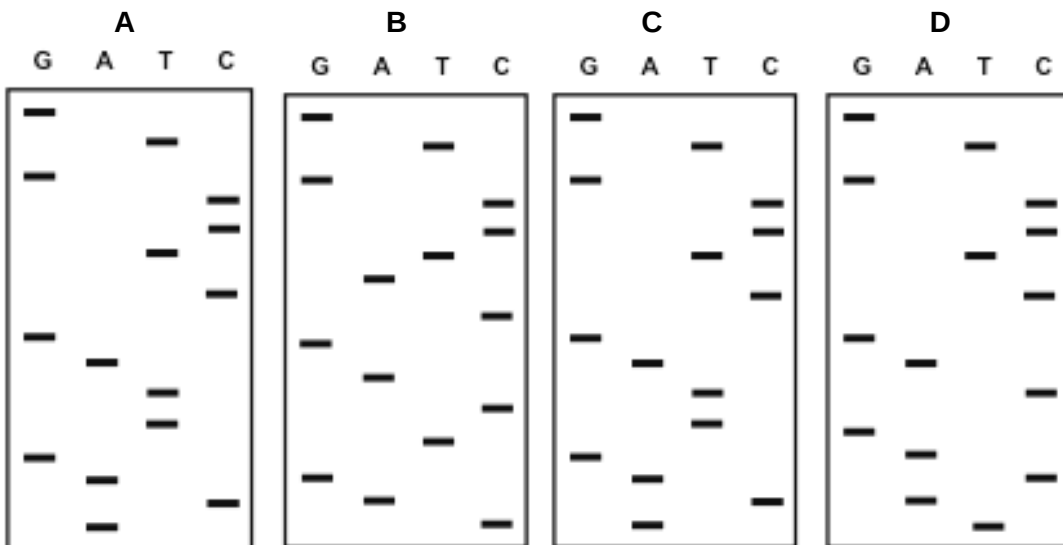
	Individual 2	Individual 4
A	The protein has an additional negatively charged amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
B	The protein has an additional positively charged amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
C	The protein has an additional negatively charged amino acid.	A non-charged amino acid in the protein has been replaced by a positively charged amino acid.
D	The protein has an additional positively charged amino acid.	A non-charged amino acid in the protein has been replaced by a positively charged amino acid.

- 38** In people with a gene mutation where a base insertion has occurred, the protein formed has a very different primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using gel electrophoresis.

The electrophoresis results from the DNA of a normal allele of the gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



39 SDS PAGE is used primarily to separate

- A** DNA based on shape
- B** Blood serum based on size

- C** RNA based on charge
- D** Channel proteins based on charge

40 Embryonic stem cells offer great promise to the cure of many diseases because

- (i) they are pluripotent.
- (ii) they are more readily available than other cells.
- (iii) they can be easily cultured in vitro.
- (iv) they will differentiate and produce cells which are normal.

Which of the following is/are true?

- A** ii, iii, iv
- B** i, iii, iv
- C** i, iv
- D** iv only